Mathematical Proposition and Proof: The Bloch-Floquet Winding Quotient Map

Proposition (Winding Quotient Homomorphism)

Let $\mathcal{H}_C^{\text{univ}} = L^2(\mathbb{R}, dt)$ be the universal cover clock Hilbert space endowed with self-adjoint momentum operator $\widehat{P}_t = -i\hbar \frac{d}{dt}$ and generalized eigenstates $\vert{}t\rangle$ satisfying $\langle t \vert{} t' \rangle = \delta(t - t')$. Let $\mathcal{H}_C^{\text{circ}} = H^2(S^1)$ be the Hardy space clock Hilbert space on the circle $S^1 \cong \mathbb{R}/(T_0\mathbb{Z})$ of fundamental period $T_0 = \frac{2\pi}{\omega}$, with orthonormal number basis $\{\vert{}n\rangle\}_{n=0}^\infty$ and generator $\widehat{N}$ such that $\widehat{H}_C = \hbar\omega \widehat{N}$.
There exists a continuous, isometric chain-map embedding and quotient projection (the Bloch–Floquet winding map) $\Pi_\omega : L^2(\mathbb{R}) \to L^2(S^1)$ defined by periodic wrapping, such that the universal cover constraint operator $\widehat{C}_{\text{univ}} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S$ maps equivariantly under Π_ω to the compact circular constraint operator $\widehat{C}_{\text{circ}} = \hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S$, and the universal cover generalized phase states $\vert{}t\rangle$ push forward to the covariant Holevo phase POVM elements $\vert{}\theta)(\theta\vert{}$ on S^1.

Proof

1. Construction of the Winding Quotient Map

Define the fundamental period $T_0 = \frac{2\pi}{\omega}$. For any function $f(t) \in \mathcal{S}(\mathbb{R})$ in the Schwartz space, its periodic periodization (the Bloch–Floquet transform / winding quotient map) $\Pi_\omega f : S^1 \to \mathbb{C}$ is given by the convergent sum over winding sectors $m \in \mathbb{Z}$:
$$(\Pi_\omega f)(\theta) = \sum_{m=-\infty}^{\infty} f\left(\frac{\theta + 2\pi m}{\omega}\right),$$
where $\theta \in [0, 2\pi)$. By Poisson summation, this map intertwines the translation operator on $\mathbb{R}$ with rotations on S^1.
The adjoint map $\Pi_\omega^* : L^2(S^1) \to L^2(\mathbb{R})$ lifts functions on the circle back to the universal cover:
$$(\Pi_\omega^* g)(t) = g(\omega t \pmod{2\pi}) \cdot w(t),$$
where $w(t)$ is a smooth partition of unity window function of compact support width T_0 ensuring normalizability across $\mathbb{R}$.

2. Push-Forward of the Constraint Operator

On the universal cover $\mathcal{H}_C^{\text{univ}} = L^2(\mathbb{R}, dt)$, the constraint operator is:
$$\widehat{C}_{\text{univ}} = -i\hbar \frac{d}{dt} \otimes I_S + I_C \otimes \widehat{H}_S.$$
A universal cover state $\vert{}\Psi_{\text{univ}}\rangle \in L^2(\mathbb{R}) \otimes \mathcal{H}_S$ satisfies $\widehat{C}_{\text{univ}} \vert{}\Psi_{\text{univ}}\rangle = 0$, which yields the exact

evolution equation:
$$-i\hbar \frac{\partial}{\partial t} \vert{}\Psi_{\text{univ}}(t)\rangle + \widehat{H}_S \vert{}\Psi_{\text{univ}}(t)\rangle = 0 \implies i\hbar \frac{\partial}{\partial t} \vert{}\Psi_{\text{univ}}(t)\rangle = \widehat{H}_S \vert{}\Psi_{\text{univ}}(t)\rangle.$$
Applying the winding quotient map $\Pi_\omega \otimes I_S$ to $\vert{}\Psi_{\text{univ}}(t)\rangle$ via the chain rule with $\theta = \omega t$:
$$\frac{\partial}{\partial t} = \omega \frac{\partial}{\partial \theta},$$
transforms Eq. (4) into:
$$i\hbar \omega \frac{\partial}{\partial \theta} \vert{}\Psi_{\text{circ}}(\theta)\rangle = \widehat{H}_S \vert{}\Psi_{\text{circ}}(\theta)\rangle.$$
Expanding the circular state in the number basis of $H^2(S^1)$, where $\widehat{N} = -i \frac{\partial}{\partial \theta}$ acts as the number operator with integer spectrum $n \ge 0$, Eq. (6) becomes:
$$\left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right) \vert{}\Psi_{\text{circ}}\rangle = \widehat{C}_{\text{circ}} \vert{}\Psi_{\text{circ}}\rangle = 0,$$
thereby proving that the universal cover constraint $\widehat{C}_{\text{univ}}$ pushes forward exactly to the compact circular constraint $\widehat{C}_{\text{circ}}$ under the winding quotient Π_ω.

3. Push-Forward of the Generalized Phase States

On the universal cover, the clock eigenstates are $\vert{}t\rangle$ with inner product $\langle t \vert{}\, t' \rangle = \delta(t - t')$. Under the winding quotient, a point $t \in \mathbb{R}$ maps to the equivalence class $\theta = \omega t \pmod{2\pi}$ combined with its winding number $m = \lfloor \omega t / 2\pi \rfloor$.
Summing over all universal cover time states $\vert{}t + \frac{2\pi m}{\omega}\rangle$ in each winding sector yields the periodic Dirac comb distribution on $\mathbb{R}$, which pulls back under Π_ω to the generalized Hardy space phase states:
$$\vert{}\theta) = \sum_{n=0}^{\infty} e^{in\theta} \vert{}n\rangle = \sum_{m=-\infty}^{\infty} \left\vert{}\, t = \frac{\theta + 2\pi m}{\omega} \right\rangle.$$
Consequently, the continuous universal cover projection measure $\vert{}t\rangle\langle t\vert{}\, dt$ wraps under Π_ω into the covariant Holevo phase POVM elements:
$$E(d\theta) = \frac{1}{2\pi} \vert{}\theta)(\theta\vert{}\, d\theta = \left(\sum_{m=-\infty}^{\infty} \left\vert{}\, \frac{\theta + 2\pi m}{\omega} \right\rangle \left\langle \frac{\theta + 2\pi m}{\omega} \right\vert{}\, \right) \frac{d\theta}{2\pi}.$$
Thus, the singular normalization structure on the universal cover $\mathbb{R}$ is smoothly compacted into the positive-operator-valued measure $E(d\theta)$ on S^1 without spectral contradiction, bridging Resolution A and Resolution B. $\blacksquare$